

Lanka Sree Naina

B.Tech - Computer Science and Engineering

Vellore Institute of Technology (VIT), Chennai, India

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EDUCATION

Vellore Institute of Technology, Chennai | *B.Tech, CSE Core* | **Aug 2023 – May 2027** CGPA: 8.61/10.

Narayana Junior College, Proddatur | *MPC* | **Jun 2021 – Apr 2023** Score: 970/1000.

Narayana Olympiad School, Proddatur | *SSC* | **Jul 2019 – May 2020** Score: 582/600.

TECHNICAL SKILLS

- **Languages:** Java, Python, JavaScript, SQL, C
- **Frameworks & Technologies:** Spring Boot, React, Node.js, Express.js, MongoDB, AWS, Docker, Pandas, NumPy, PyTorch
- **Core Concepts:** Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), DBMS, Operating Systems, Computer Networks
- **Tools & Platforms:** Git, GitHub, VS Code, Google Colab

EXPERIENCE & PROJECTS

DentalXNet: AI-Powered OPG Analysis System | [Github Link](#) | Python, PyTorch, FastAPI, Next.js, RT-DETR, YOLOv8

Developed a full-stack AI-powered dental radiograph analysis platform for automated detection and localisation of crowns, implants, bridges, root-canal-treated teeth, dental fillings, and endodontic posts from panoramic OPG X-rays.

- Designed and benchmarked CNN-based YOLOv8 and Transformer-based RT-DETR architectures on 2,235 annotated dental radiographs, achieving **85.3% Precision** and **71.3% mAP50**.
- Enhanced model interpretability through attention-based analysis, detection visualisation, and explainable AI dashboards for clinical image understanding.
- Built and deployed a complete web platform using **Next.js, FastAPI, PyTorch, Vercel**, and **Hugging Face Spaces**, featuring interactive inference, patient record management, PDF report generation, and model comparison workflows.

AI Wi-Fi Security & Risk Assessment | Python, FastAPI, React, Scikit-learn, Nmap, Isolation Forest, Random Forest

Developed an AI-assisted cybersecurity prototype to analyse Wi-Fi metadata, detect network anomalies, and assess security risks using machine learning.

- Implemented a hybrid risk assessment engine combining Isolation Forest anomaly detection with rule-based vulnerability analysis.
- Built a FastAPI backend and React dashboard for real-time network monitoring, telemetry visualisation, and risk reporting.
- Integrated Nmap and system-level network utilities to identify connected devices, detect open ports, and evaluate potential security misconfigurations.

Course Feedback System | React, Node.js, Express, MongoDB, JWT

Developed a full-stack academic feedback platform with role-based access control.

- Implemented JWT-based authentication and authorisation to secure access across multiple user roles.
- Built feedback workflows enabling submission, review, and response management through role-specific dashboards.
- Integrated a React frontend with a Node.js/Express backend and MongoDB for efficient data storage and retrieval.

College Feedback Classifier | Python, NLP, Hugging Face, BART, Gradio, CLI

Built an intelligent feedback classification system using zero-shot NLP techniques.

- Applied the BART model via Hugging Face to categorise feedback into domains like Academics, Facilities, and Administration.
- Developed both CLI and Gradio web interfaces supporting CSV input/output and real-time classification.
- Added visual analytics with dynamic charts for insights and decision-making.

RESEARCH

Summer Research Internship Program, VIT Chennai

- Explored Knowledge Graph concepts and Neo4j graph database fundamentals as part of a faculty-guided research project.

ACHIEVEMENTS & CERTIFICATIONS

- AWS Academy Graduate – AWS Academy Cloud Foundations (2026)
- Udemy Full Stack Web Development (2026)
- IBM Generative AI Explorer Certificate (2025)
- Salesforce Agentblazer Champion, Trailhead Achievement (2025)

EXTRACURRICULAR ACTIVITIES

- Represented Andhra Pradesh in cultural and international events through Kuchipudi dance at VIT Chennai; served as coordinator for Sirivennela TLA, contributing to event organization and cultural activities.

Languages

Telugu (Native), English (Fluent), Hindi (Intermediate), Tamil (Basic), French (Basic)